

Time-Series Forecasting

10 kHz vibration sensors. An LSTM + anomaly detector flags bearing failure 72 hours ahead. Unplanned downtime dropped 41%. Today we build the forecasting core.

What is the one time-series you personally produce every day?

WHAT'S INSIDE

- 01** Why time-series is different
- 02** Decomposition
- 03** Three forecasters — when to pick
- 04** Lag + rolling features
- 05** From forecast to alert

WHY THIS LESSON MATTERS

Reliance Jamnagar: ₹30 crore saved per refinery day.

NUMBER 1

10 k

From real-world deployments.

NUMBER 2

41%

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Decompose a series into trend + seasonality + residual.

2

Choose between ARIMA, Prophet, and LSTM.

3

Engineer lag and rolling-window features.

4

Forecast with confidence intervals.

5

Integrate a forecast into a predictive-maintenance alert.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Why time-series is different

Leakage and splits

02

Decomposition

Trend + seasonality + residual

03

Three forecasters — when to...

Gradient-boosted trees on lag features often beat LSTMs at small scale.

04

Lag + rolling features

Cheap, powerful, interpretable

01

PART 1 OF 4

Why time-series is different

Leakage and splits

CONCEPT 1 OF 4

Why time-series is different

Rows are NOT independent.

Train = past; test = future. Never random-split.

Metrics must respect temporal order.

Feature engineering uses only past values.

CARDINAL RULE

Never random-split

Random splits leak the future into the training set.

DEFINITION

Why time-series is different

Leakage and splits

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Decomposition

Trend — long-run direction.

Seasonality — repeating pattern (weekly, yearly).

Residual — unexplained; what the model must capture.

`statsmodels.seasonal_decompose` in 2 lines.

FIRST STEP

Plot all three

The shape dictates the model.

KEY POINTS

- Trend — long-run direction
- Seasonality — repeating pattern (weekly, yearly)
- Residual — unexplained; what the model must capture
- `statsmodels.seasonal_decompose` in 2 lines

CONCEPT 3 OF 4

Three forecasters — when to pick

Gradient-boosted trees on lag features often beat LSTMs at small scale.

DEFINITION

Three forecasters — when to pick

Gradient-boosted trees on lag features often beat LSTMs at small scale.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Lag + rolling features

```
df["lag_1"] = df["value"].shift(1)  
df["lag_7"] = df["value"].shift(7)  
df["roll_30_mean"] = df["value"].rolling(30).mean()
```

Feed into LightGBM — fast to train, easy to debug.

RULE

Lags only use past

Compute on a time-ordered frame; never shuffle before.

KEY POINTS

- `df["lag_1"] = df["value"].shift(1)`
- `df["lag_7"] = df["value"].shift(7)`
- `df["roll_30_mean"] = df["value"].rolling(30).mean()`
- Feed into LightGBM — fast to train, easy to debug

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Decompose a real series

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where have you seen a random-split leak the future?

Q2

Which model would you start with on Flipkart BBD data?

Q3

What confidence band matters for your project?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Why never random-split a time series?

A

Slow

B

Future leaks into training

C

Bug

D

Memory

Why never random-split a time series?

CORRECT ANSWER



Future leaks into training

WHY

Evaluation becomes dishonest; the model is not actually forecasting

Daily sales typical seasonality:

A

Every 2 days

B

Weekly (7)

C

Monthly (30)

D

Annual (365)

Daily sales typical seasonality:



CORRECT ANSWER

Weekly (7)

WHY

Weekday/weekend cycle dominates most retail series

Prophet shines when:

A

Data is clean

B

Data has holidays + missing values

C

Data is tiny

D

Data is i.i.d.

Prophet shines when:



CORRECT ANSWER

Data has holidays + missing values

WHY

Prophet's additive model handles holidays and gaps gracefully

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Rows are NOT independent.

Train = past; test = future. Never random-split.

Metrics must respect temporal order.

Feature engineering uses only past values.

CARDINAL RULE

Never random-split

Random splits leak the future into the training set.

THE TAKEAWAY

"Those habits survive production. The model is negotiable; the habits are not."

WHAT TO NOTICE

1

Why time-series is different

2

Decomposition

3

Three forecasters — when to...

4

Lag + rolling features

Mini assignment — forecast Bangalore AQI

Use Prophet or LightGBM + lag features to forecast daily AQI for 7 days.

DELIVERABLES

- Download Bangalore AQI history.
- Train on 80% (temporally earliest).
- Forecast the next 7 days with a 95% CI.
- Post forecast plot + 3-sentence interpretation.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Forecast plot + short interpretation on the forum.

ONE THING TO REMEMBER

“

*"Those habits survive production. The model is negotiable;
the habits are not."*

— AI Learning Hub

END OF LESSON 4

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 5 — React Fundamentals. Bind your prediction API to a UI.